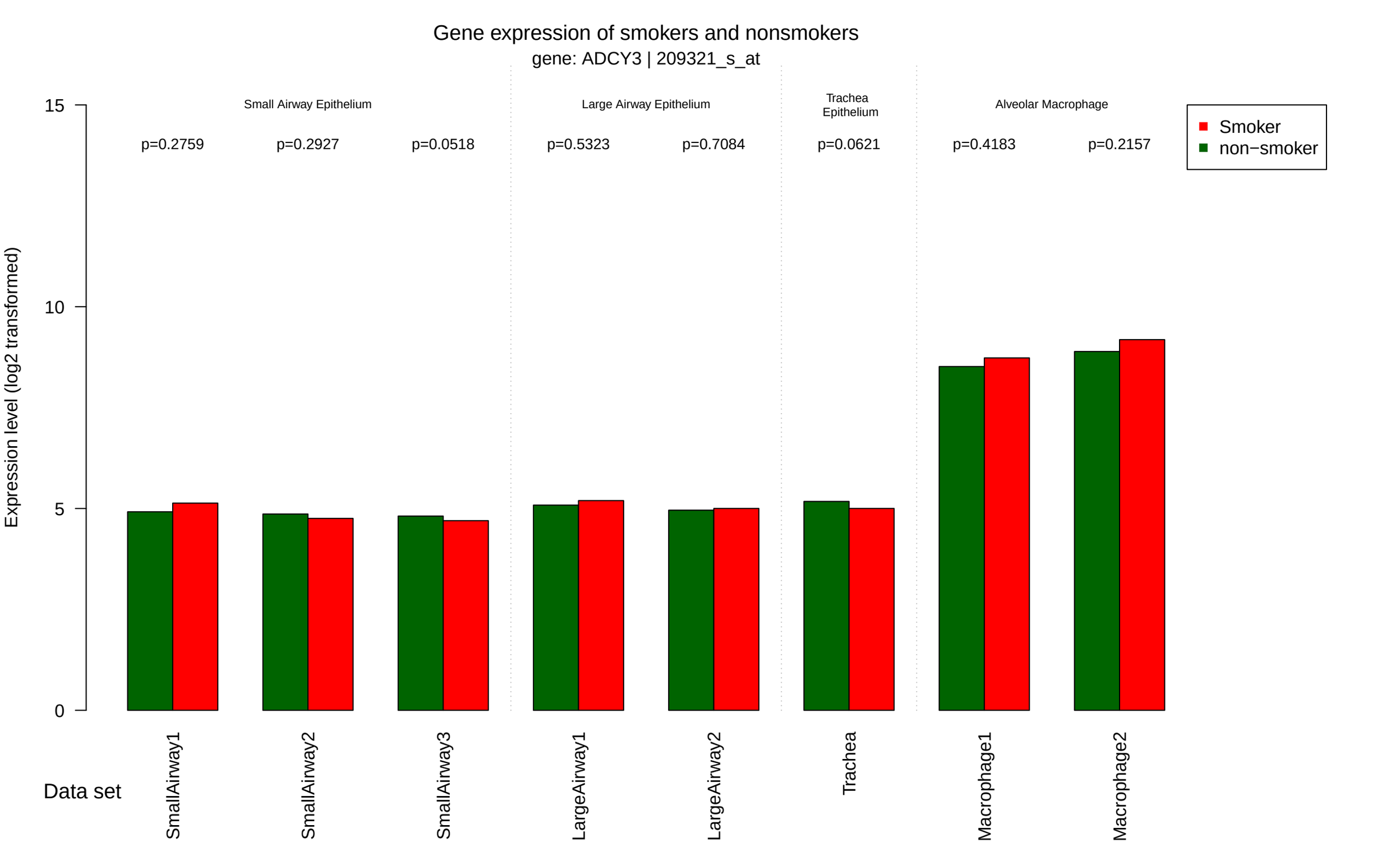
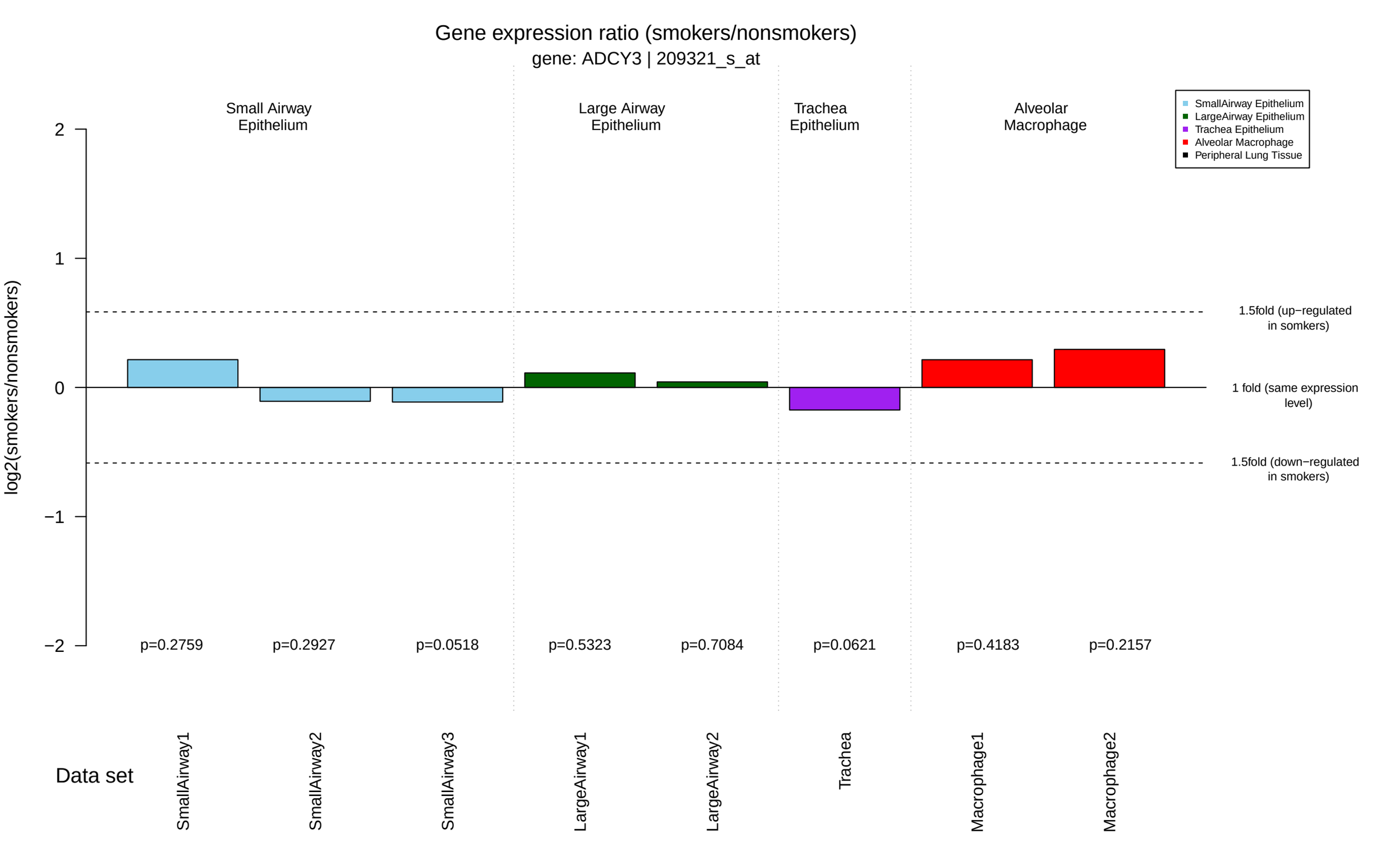
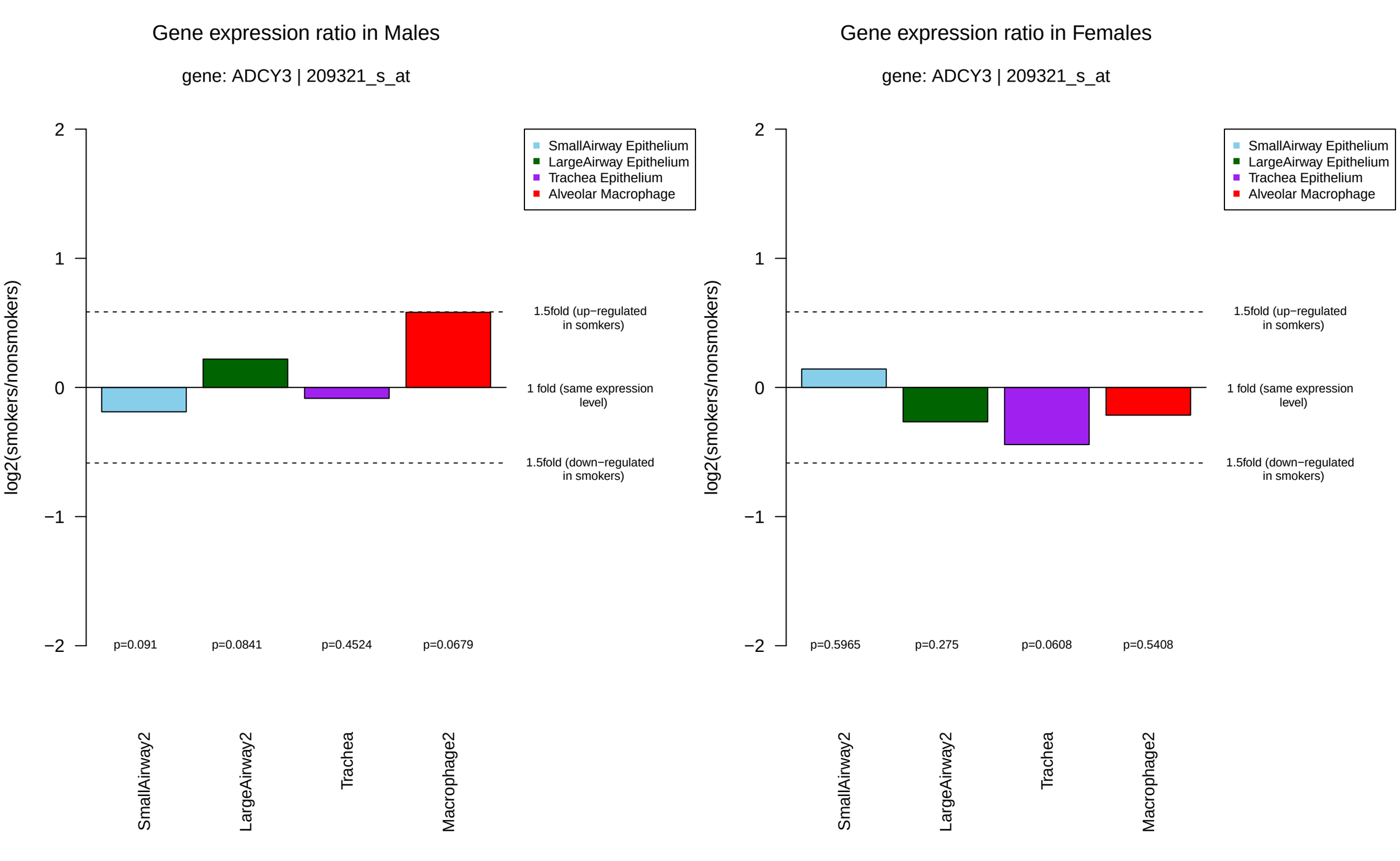
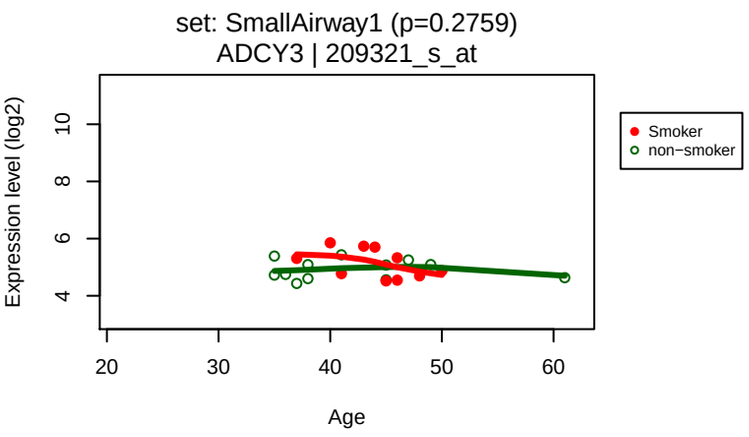
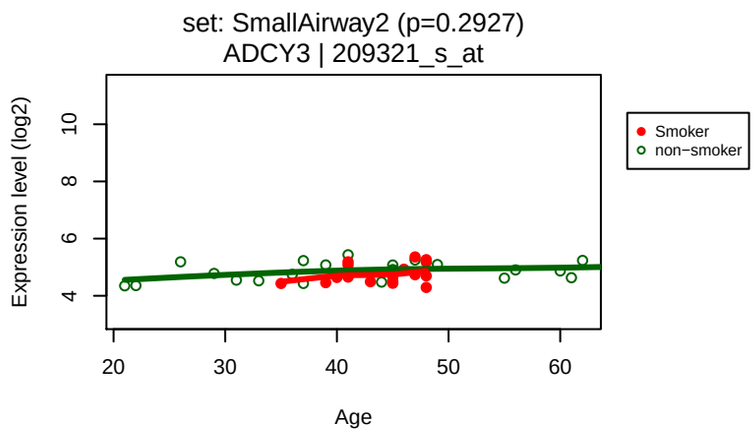
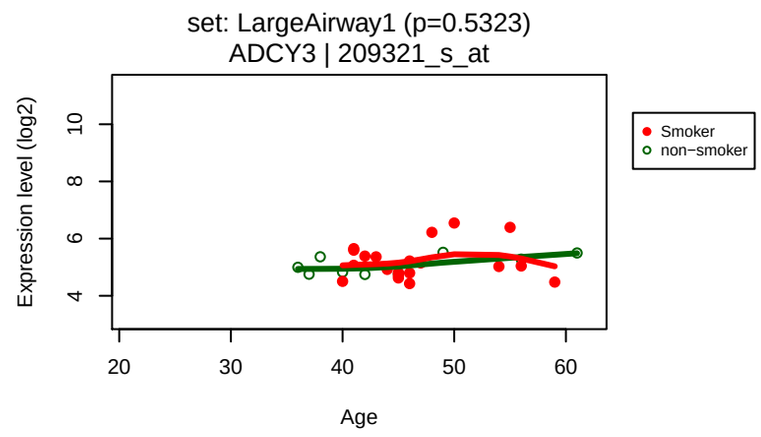

Gene expression of smokers and nonsmokers
gene: ADCY3 | 209321_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2759
p=0.2927
p=0.0518
p=0.5323
p=0.7084
p=0.0621
p=0.4183
p=0.2157
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

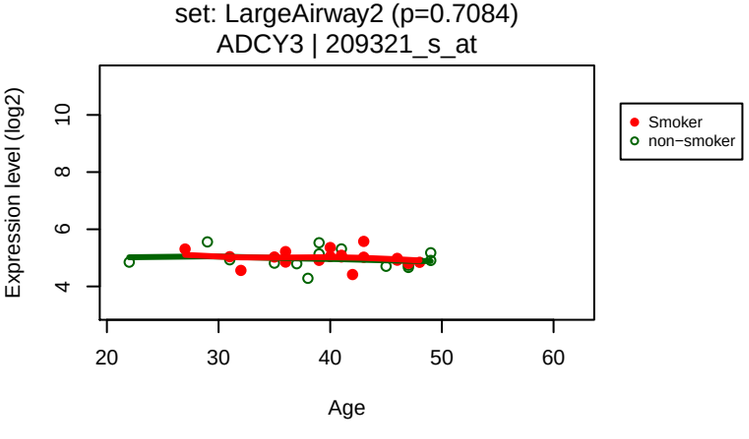

Gene expression ratio (smokers/nonsmokers)
gene: ADCY3 | 209321_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2759
p=0.2927
p=0.0518
p=0.5323
p=0.7084
p=0.0621
p=0.4183
p=0.2157
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

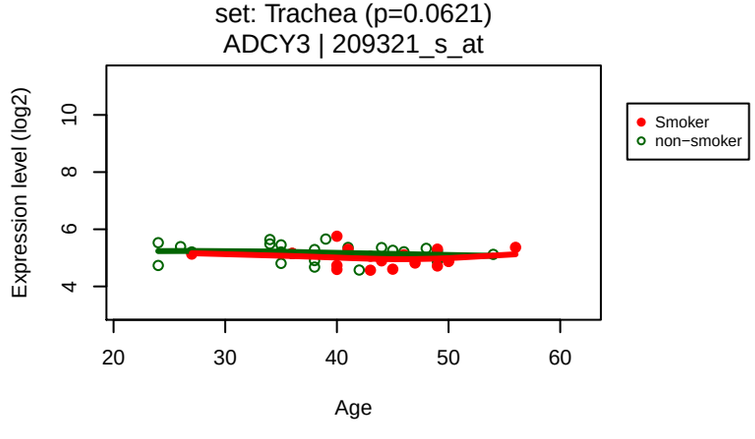

Gene expression ratio in Males
gene: ADCY3 | 209321_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.091
p=0.0841
p=0.4524
p=0.0679
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCY3 | 209321_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5965
p=0.275
p=0.0608
p=0.5408
SmallAirway2
LargeAirway2
Trachea
Macrophage2

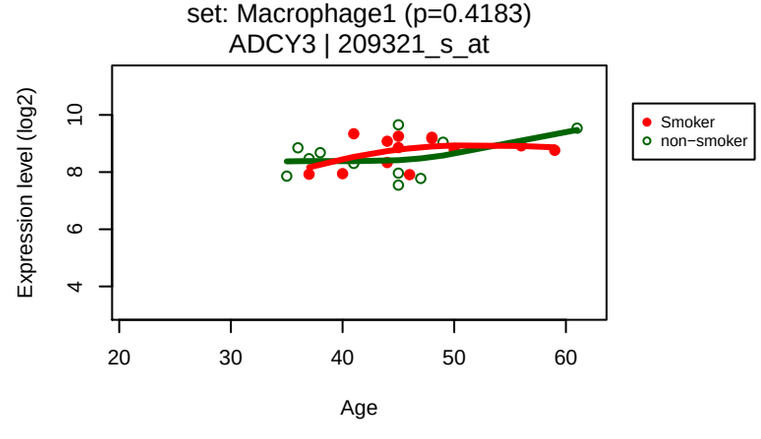

set: SmallAirway1 (p=0.2759)
ADCY3 | 209321_s_at
Expression level (log2)
Age
Smoker
non−smoker

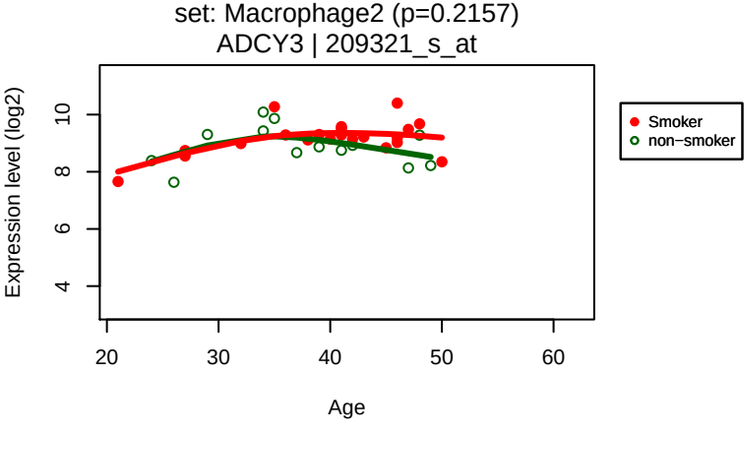

set: SmallAirway2 (p=0.2927)
ADCY3 | 209321_s_at
Expression level (log2)
Age
Smoker
non−smoker

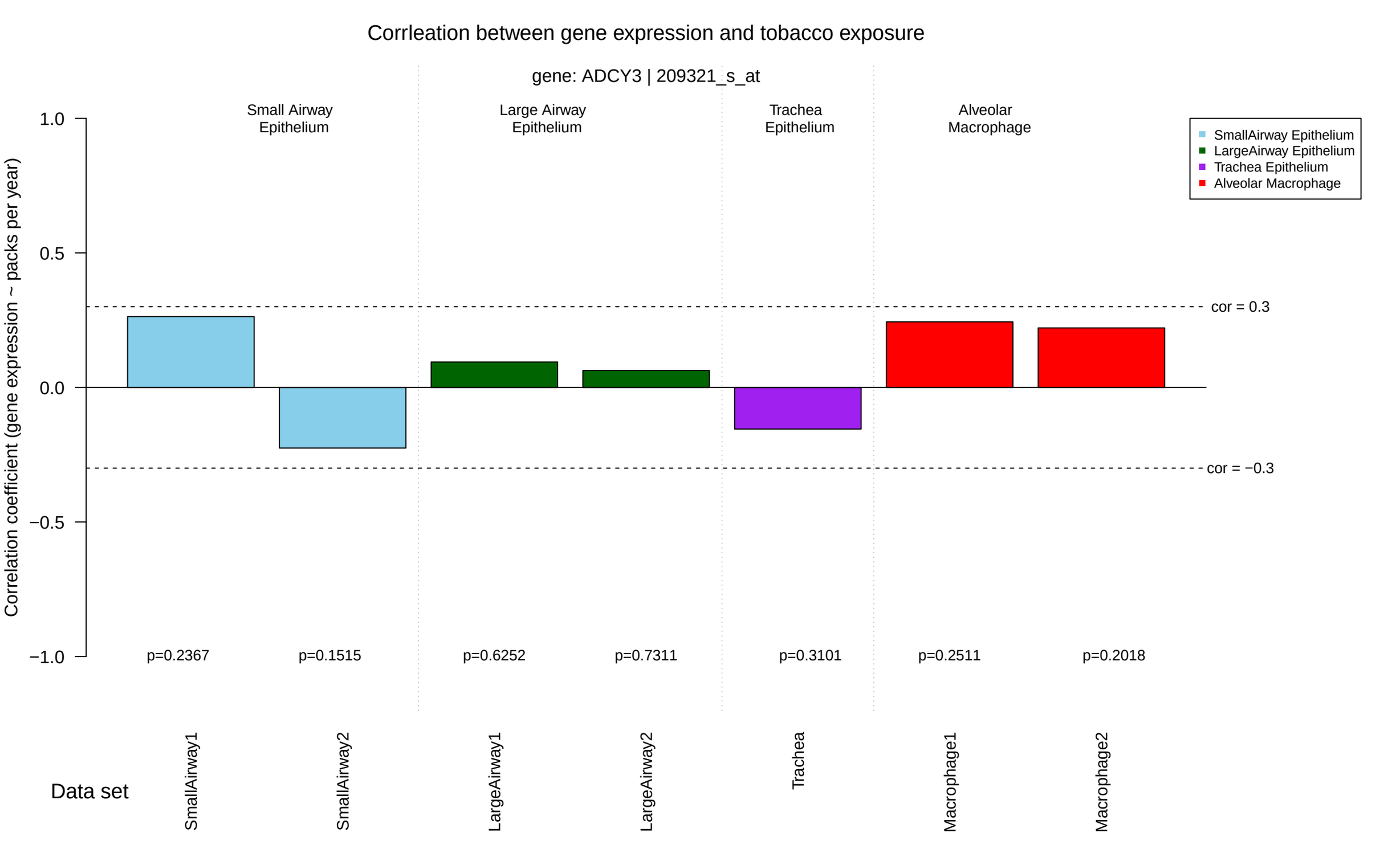

set: LargeAirway1 (p=0.5323)
ADCY3 | 209321_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7084)
ADCY3 | 209321_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0621)
ADCY3 | 209321_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4183)
ADCY3 | 209321_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2157)
ADCY3 | 209321_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCY3 | 209321_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2367
p=0.1515
p=0.6252
p=0.7311
p=0.3101
p=0.2511
p=0.2018
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2